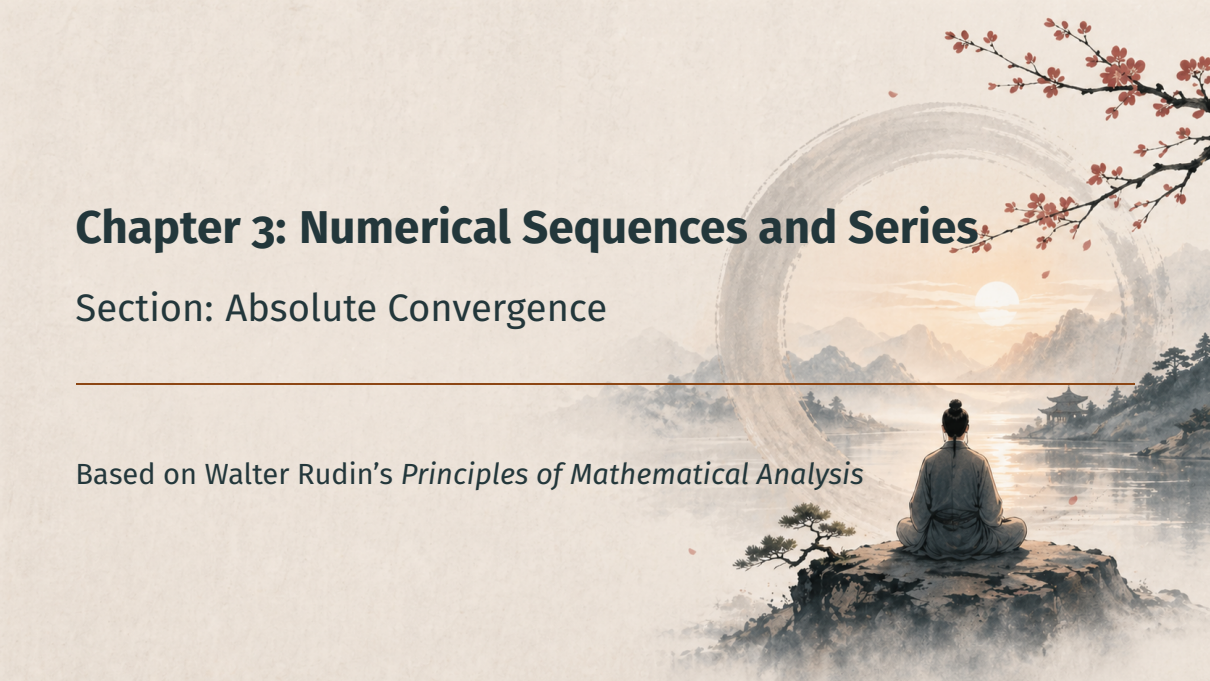


Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person with their hair in a bun, wearing a grey robe, sits in a meditative lotus position on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains rise. A large, faint, circular frame, resembling a ring or a portal, is centered in the background, with a bright sun or moon setting or rising within it. A branch with red blossoms hangs from the top right corner. A small pavilion is visible on the right side of the lake. A thin horizontal line is drawn across the middle of the image, separating the title section from the subtitle section.

Section: Absolute Convergence

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

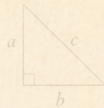
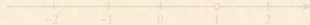
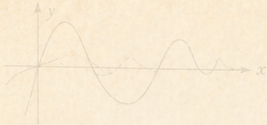
Definition

Main Result

Proof

Remarks and Consequences

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why Absolute Convergence?

- Given $\sum a_n$, also look at the series of sizes $\sum |a_n|$



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Why Absolute Convergence?

- Given $\sum a_n$, also look at the series of **sizes** $\sum |a_n|$
- Cancellation can save a series:
 $\sum \frac{(-1)^n}{n}$ converges, but $\sum \frac{1}{n}$ **diverges**



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why Absolute Convergence?

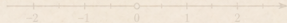
- Given $\sum a_n$, also look at the series of **sizes** $\sum |a_n|$
- Cancellation can save a series:
 $\sum \frac{(-1)^n}{n}$ converges, but $\sum \frac{1}{n}$ **diverges**
- **Question:** when does convergence **not** rely on cancellation?



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



$$x^2 + y^2 = r^2$$

Definition



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Absolute Convergence

Definition: Absolute Convergence

The series $\sum a_n$ is said to **converge absolutely** if the series $\sum |a_n|$ converges.

Note: the terms a_n may be *real or complex*; $|a_n|$ is the absolute value (or modulus).

Main Result



Theorem 3.45

Theorem 3.45

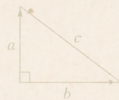
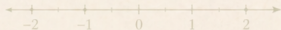
If $\sum a_n$ converges **absolutely**, then $\sum a_n$ **converges**.

$\sum |a_n|$ converges
(absolute)

always

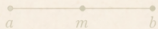
$\sum a_n$ converges

not in general



$$a^2 + b^2 = c^2$$

Proof



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Proof Strategy: Theorem 3.45

Goal: from convergence of $\sum |a_n|$, deduce convergence of $\sum a_n$.

Tools:

1. The **triangle inequality** for finite sums.
2. The **Cauchy criterion** for series (Theorem 3.22).

Proof — Step 1: Recall the Cauchy Criterion

Recall: Theorem 3.22 (Cauchy Criterion)

$\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer N such that

$$\left| \sum_{k=n}^m a_k \right| \leq \varepsilon \quad (m \geq n \geq N).$$

Proof — Step 2: The Key Inequality

Triangle inequality for finite sums

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k|.$$

The size of a block of Σa_n is controlled by the corresponding block of $\Sigma |a_n|$.

Proof — Step 3: Combine and Conclude

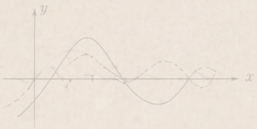
Given $\varepsilon > 0$, apply the Cauchy criterion to $\Sigma|a_n|$: there is N with

$$\sum_{k=n}^m |a_k| \leq \varepsilon \quad (m \geq n \geq N).$$

Then, by Step 2,

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k| \leq \varepsilon \quad (m \geq n \geq N).$$

So Σa_n satisfies the Cauchy criterion. Hence it converges.

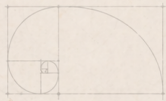


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Remarks and Consequences



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

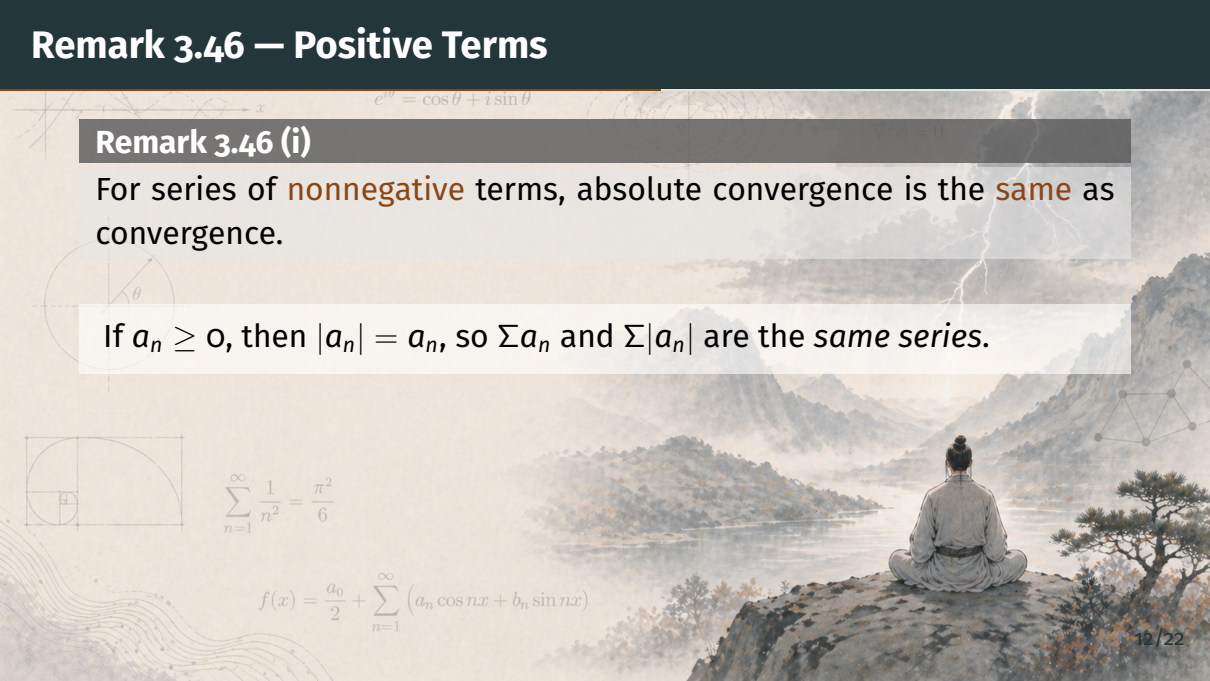
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Positive Terms

Remark 3.46 (i)

For series of **nonnegative** terms, absolute convergence is the **same** as convergence.

If $a_n \geq 0$, then $|a_n| = a_n$, so $\sum a_n$ and $\sum |a_n|$ are the *same series*.


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

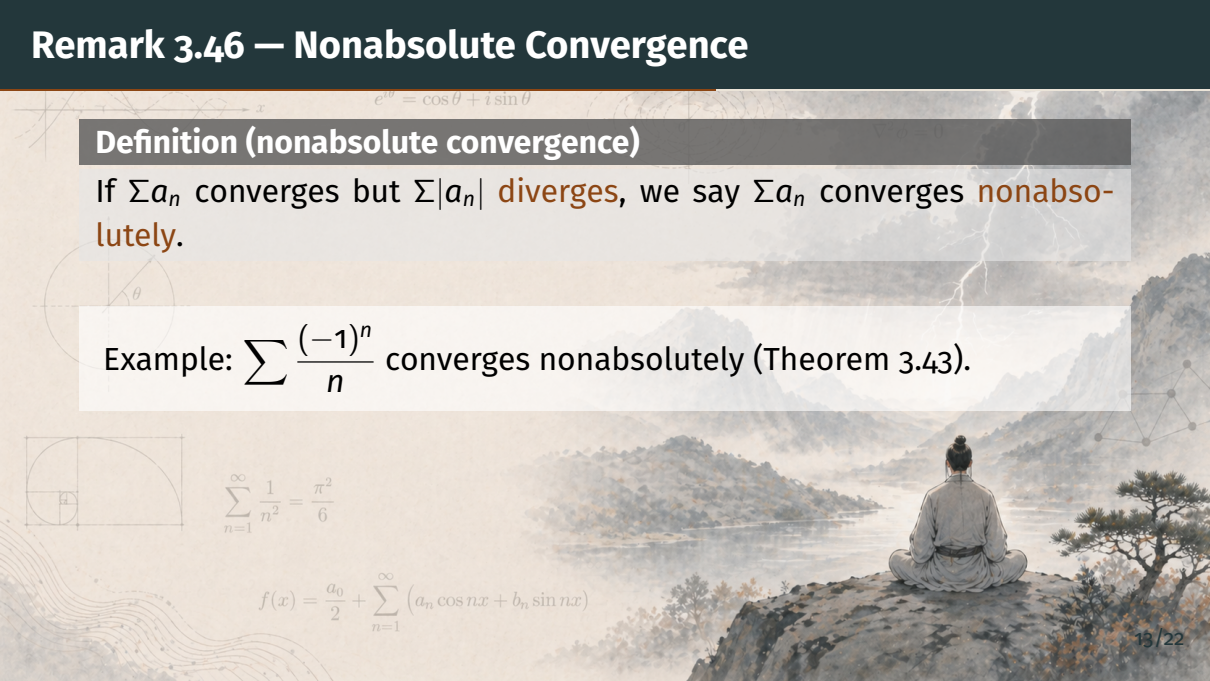
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Nonabsolute Convergence

Definition (nonabsolute convergence)

If $\sum a_n$ converges but $\sum |a_n|$ **diverges**, we say $\sum a_n$ converges **nonabsolutely**.

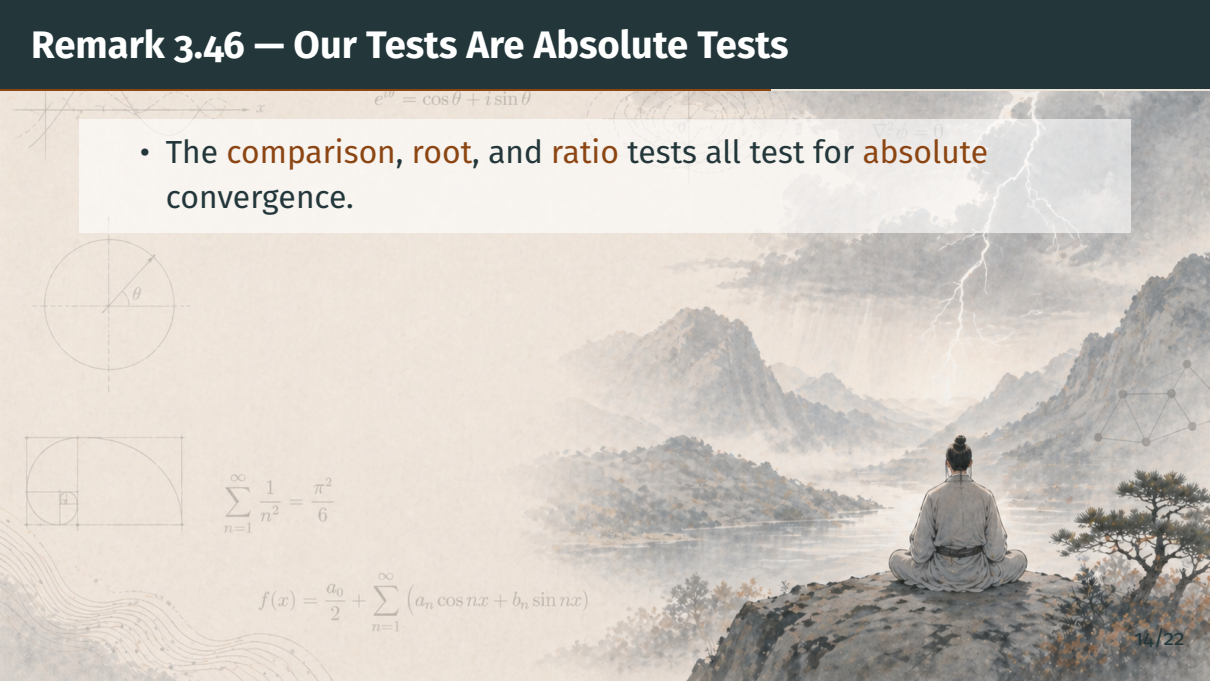
Example: $\sum \frac{(-1)^n}{n}$ converges nonabsolutely (Theorem 3.43).


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Our Tests Are Absolute Tests

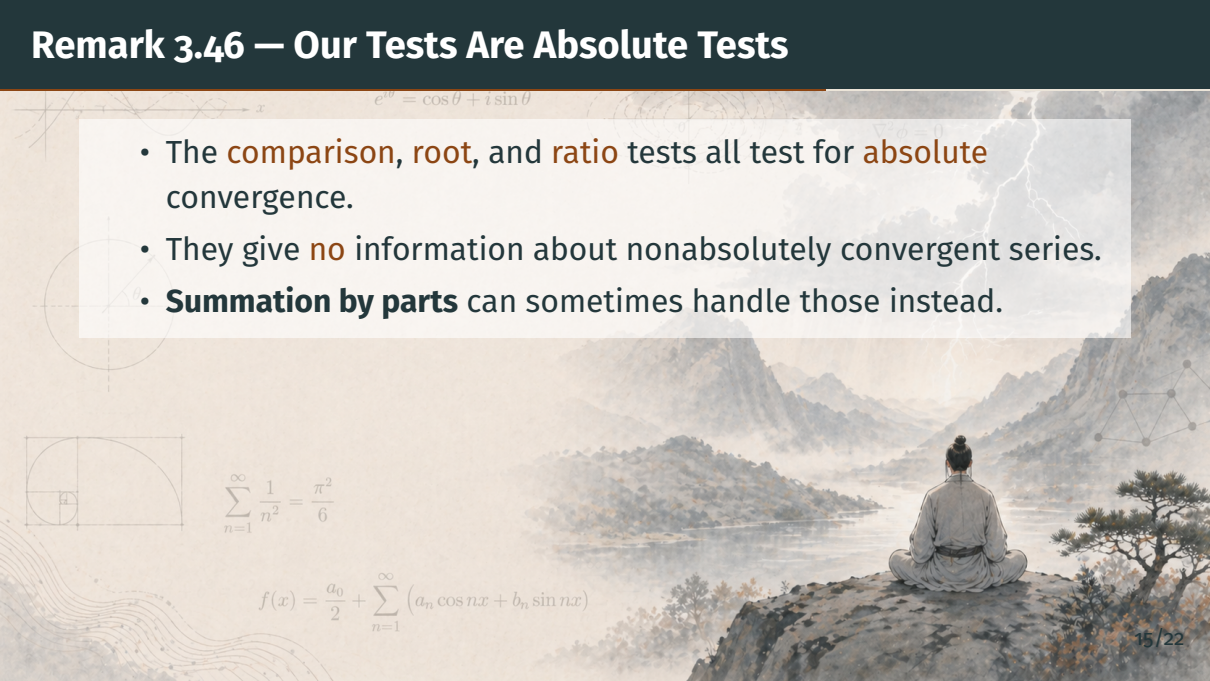
- The **comparison**, **root**, and **ratio** tests all test for **absolute** convergence.

The background of the slide is a collage of various elements. At the top, there's a dark blue header with the title. Below it, a light beige background features several faint, artistic illustrations: a sine wave on the left, a unit circle with an angle theta in the middle-left, a square with an inscribed circle and a smaller square inside it on the bottom-left, and a network graph on the right. A large, traditional Chinese ink wash painting of a person meditating on a cliff overlooking a misty mountain landscape with a lightning bolt in the sky occupies the right half. Mathematical formulas are scattered throughout: $e^{i\theta} = \cos \theta + i \sin \theta$ at the top center, $\nabla^2 \phi = 0$ at the top right, $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ in the middle-left, and $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$ at the bottom center.
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Our Tests Are Absolute Tests

- The **comparison**, **root**, and **ratio** tests all test for **absolute** convergence.
- They give **no** information about nonabsolutely convergent series.
- **Summation by parts** can sometimes handle those instead.

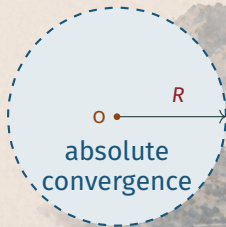

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Power Series

Remark 3.46 (continued)

A power series converges **absolutely** in the **interior** of its circle of convergence.



boundary $|z| = R$: separate question

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Why It Matters

Absolutely convergent series behave like finite sums

We may **rearrange** the terms and **multiply** two such series term by term, without changing the sum.

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

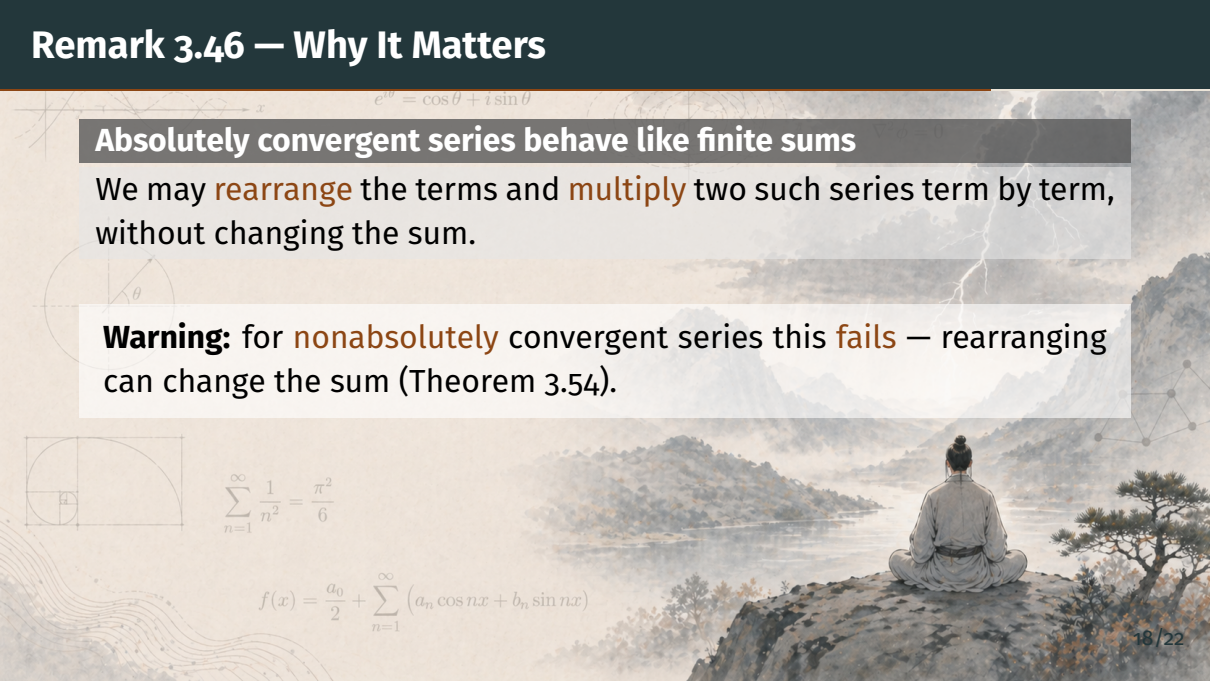
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark 3.46 — Why It Matters

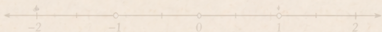
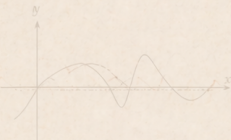
Absolutely convergent series behave like finite sums

We may **rearrange** the terms and **multiply** two such series term by term, without changing the sum.

Warning: for **nonabsolutely** convergent series this **fails** — rearranging can change the sum (Theorem 3.54).

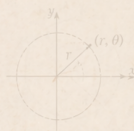

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



$$a^2 + b^2 = c^2$$

Summary



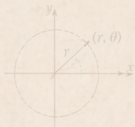
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways from this section:

- **Definition:** $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways from this section:

- **Definition:** $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.
- **Theorem 3.45:** absolute convergence $\Rightarrow$ convergence, via the triangle inequality plus the Cauchy criterion.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways from this section:

- **Definition:** $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.
- **Theorem 3.45:** absolute convergence $\Rightarrow$ convergence, via the triangle inequality plus the Cauchy criterion.
- **Converse fails:** $\sum (-1)^n/n$ converges nonabsolutely; our comparison, root, and ratio tests are absolute tests.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways from this section:

- **Definition:** $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.
- **Theorem 3.45:** absolute convergence $\Rightarrow$ convergence, via the triangle inequality plus the Cauchy criterion.
- **Converse fails:** $\sum (-1)^n/n$ converges nonabsolutely; our comparison, root, and ratio tests are absolute tests.
- **Payoff:** absolutely convergent series can be rearranged and multiplied like finite sums.

Coming up next: addition and multiplication of series.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

